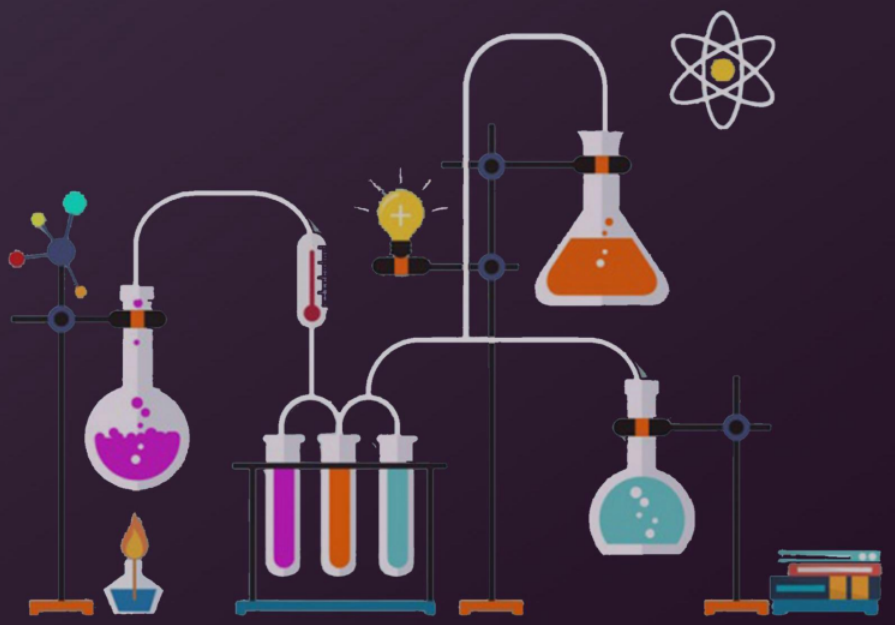
CHEMISTRY

Metallurgy



SHEET

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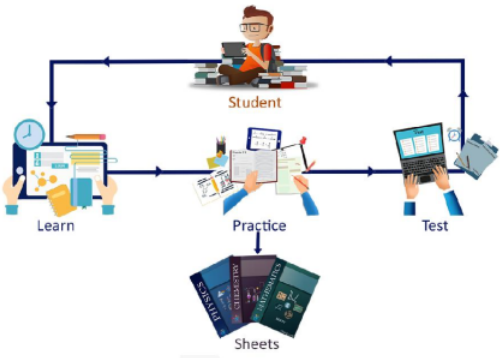
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Minerals and Ores	1			
2	Levigation, Magnetic Separation and Froth Flotation	2-6	30	1,2	1,33
3	Chemical Separation Methods	7-10	31,32	3	2
4	Roasting and Calcination	11-16	1,23		
5	Reduction Processes	17-23	2,33		3,14-16
6	Refining by Poling, Liquation and Distillation	24-25			
7	Vapour Phase Refining, Zone Refining, Pyrometallurgical Refining, Amalgamation and Electrolytic Refining	26-27	3		
8	Ellingham Diagram	28-32	4,5,6	4,5	4,5
9	Extraction of Gold, Silver and Zinc	33-38	7-8	6-9	6
10	Extraction of Tin and Copper	39-49	9-10,24	10-18	34
11	Extraction of Lead and Magnesium	50-51	25	19,20	7,8,17-20, 24-25
12	Extraction of Iron	52-54	11,12,28, 29,34	21-28	9,32
13	Extraction of Aluminium	55-60	13-22, 26,27	29-35	10,11,12,13, 21-23,26-31

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Minerals and Ores

Q1. A mineral is called ore if :
(A) The metal present in the mineral is costly
(B) A metal can be extracted from it
(C) A metal can be extracted profitably from it
(D) A metal cannot be extracted from it.

Levigation, Magnetic Separation and Froth Flotation

Q2. The ores that are concentrated by floatation method are-
(A) Carbonates (B) Sulphides
(C) Oxides (D) Phosphates

Q3. Froth flotation process is based on-
(A) Specific gravity of the ore particles
(B) Magnetic properties of the ore particles
(C) Wetting properties of the ore particles
(D) Electrical properties of the ore particles

Q4. Magnetic separation is used for increasing concentration of the following-
(A) Horn silver (B) Calcite
(C) Hamatite (D) Magnesite

Q5. Sulphide ores are generally concentrated by:-
(A) froth floatation
(B) roasting
(C) magnetic separation
(D) carbon reduction

Q6. Which of the following is used as a depressant in the separation of ZnS :-
(A) NaCN (B) Al_2O_3
(C) $Na[Ag(CN)_2]$ (D) Na_2S

Chemical Separation Methods

Q7. When silver chloride is treated with potassium cyanide-
(A) Silver is precipitated
(B) A complex ion is formed
(C) Double decomposition takes place
(D) No reaction takes place.

Q8. Which of the following metals is obtained by leaching the ore with dilute sodium cyanide solution?
(A) Silver (B) Titanium
(C) Vanadium (D) Zinc.

Q9. Which of the following is not the physical method of beneficiation of ore :-
(A) Hydraulic washing
(B) Froth floatation process
(C) Magnetic separation process
(D) Leaching

Q10. Which of the following chemical steps can be termed as leaching :-
(A) $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$
(B) $PbS + 2PbO \longrightarrow 3Pb + SO_2$
(C) $Al_2O_3.2H_2O \xrightarrow{\Delta} Al_2O_3 + 2H_2O$
(D) $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$

Roasting and Calcination

Q11. Zinc blend on roasting in air gives-
(A) Zinc carbonate
(B) SO_2 and ZnO
(C) ZnS and $ZnSO_4$
(D) CO_2 and ZnO

Q12. The main function of roasting is
(A) to make slag
(B) Oxidation
(C) Reduction
(D) Removal of volatile substance

Q13. Calcination is the process in which-
(A) Heating the ore in presence of air
(B) Heating the ore in presence of sulphur
(C) Heating the ore in absence of air
(D) Heating the ore in presence of chlorine

Q14. Calamine is :
(A) $ZnSO_4$ (B) Zn
(C) ZnS (D) $ZnCO_3$

Q15. Which of the following reaction shows calcination process :-
(A) $ZnS + O_2 \xrightarrow{\Delta} ZnO + CO_2$
(B) $FeO + SiO_2 \xrightarrow{\Delta} FeSiO_3$
(C) $CuFeS_2 \xrightarrow{\Delta} Cu_2S + FeS$
(D) $CuCO_3.Cu(OH)_2 \xrightarrow{\Delta} CuO + CO_2 + H_2O$

Q16. Iron pyrites ($CuFeS_2$) may be heated in air to remove the sulphur content of the ore. The process is known as :-
(A) roasting (B) smelting
(C) leaching (D) calcination

Reduction Processes

Q17. Which of the following can be extracted by the electrolytic reduction of their salts ?
(A) Ag (B) Mg and Al
(C) Ca (D) All

Q18. In the Down's process of obtaining sodium, $CaCl_2$ is added-
(A) To get high yield of sodium.
(B) To improve the quality of sodium obtained.
(C) To lower the operating temperature.
(D) To act as a reducing agent.

Q19. Aluminothermy used for on the spot welding of large iron structure is based on the fact that-
(A) As compared to iron, aluminium has greater affinity for oxygen.
(B) As compared to aluminium, iron has greater affinity for oxygen.
(C) Reaction between aluminium and oxygen is endothermic.
(D) Reaction between iron and oxygen is endothermic.

Q20. Al_2O_3 formation involves evolution of a large quantity of heat, which makes it useful in:
(A) Deoxidiser
(B) Indoor photography
(C) Cofectionary
(D) Thermite welding

Q21. In the alumino-thermite process, Al acts as-
(A) An oxidising agent
(B) A flux
(C) Solder
(D) A reducing agent

Q22. Thermite process is used to extract metals :
(A) When their oxides cannot be reduced by carbon.
(B) When their carbonates do not yield oxides by thermal decomposition.
(C) When their sulphides cannot be converted into oxides by roasting.
(D) When their melting points are very high.

Q23. In aluminothermite process aluminium acts as :-
(A) oxidant (B) flux
(C) reducing agent (D) solder

Refining By Poling, Liquation and Distillation

Q24. Poling process is used :-
(A) for the removal of Cu_2O from Cu
(B) for the removal of Al_2O_3 from Al
(C) for the removal of Fe_2O_3 from Fe
(D) in all the above

Q25. "Liquation" is based upon

(A) Boiling point difference between metal and impurities

(B) Fractional crystallisation of any constituent

(C) The difference in melting point of metal and impurity

(D) None of the above

Vapour Phase Refining, Zone Refining, Pyrometallurgical, Amalgamation and Electrolytic Refining

Q26. Vapour phase refining method is applicable on

(A) Ni (B) Ti (C) Zr (D) All

Q27. The method of zone refining of metal is based on the principle of :-

(A) greater solubility of the impurity in the molten state than in the solid.

(B) greater mobility of the pure metal than that of the impurity.

(C) higher melting point of the impurity than that of the pure metal.

(D) greater noble character of the solid metal than that of the impurity.

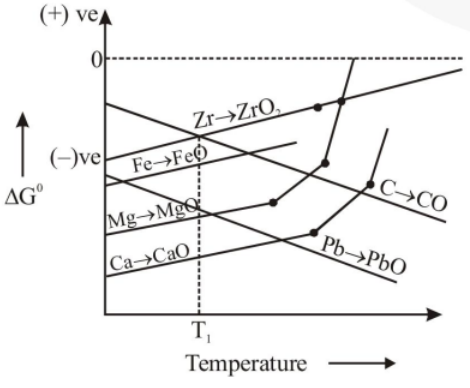
Ellingham Diagram

Q28. MgO can be reduced by C above this temperature-

(A) 1150°C (B) 1500°C

(C) 1750°C (D) 2000°C

Q29.

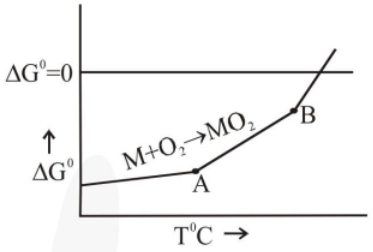


Which of the above curve is wrongly presented ?

(A) $C \rightarrow CO_2$ (B) $Pb \rightarrow PbO$

(C) $Zr \rightarrow ZrO_2$ (D) $Mg \rightarrow MgO$

Q30. The ellingham diagram of a certain metal M is represented in the figure the point A and B respectively correspond to :-



(A) Oxidation, reduction of Metal

(B) beginning and end of reduction process

(C) Minimum and maximum value of ΔG

(D) Melting point and boiling point of metal

Q31. Ellingham diagrams can be drawn for the following

(A) Sulphides

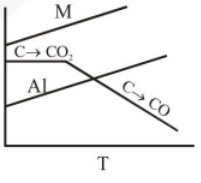
(B) oxides

(C) Halides

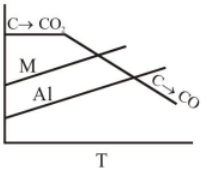
(D) All of these

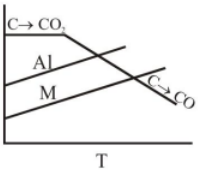
Q32. A metal (M) which exists in oxide form and its oxide is not possible to reduce by carbon up to 800°C but it is possible to reduce by Al even below 800°C. Select correct Ellingham diagram for metal (M).

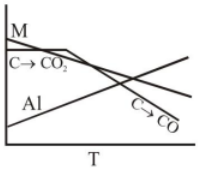
(A)



(B)



(C) 

(D) 

Extraction of Gold, Silver and Zinc

Q33. Which of the following is used to remove lead impurities from silver ?

(A) Leaching with dilute NaCN solution

(B) Parke's process

(C) Leaching with dilute NaCN solution in presence of air

(D) Solvay's process.

Q34. During extraction of silver, which of the following is formed

(A) Na $[Ag(CN)_2]$ (B) $Na_2[Ag(CN)_2]$

(C) $Na_4Ag(CN)_2$ (D) None of these

Q35. Ag is refined by cupellation process. The process removes the impurity of :

(A) Cu (B) Au

(C) Pb (D) Pt

Q36. The anode mud in the electrolytic refining of silver contains -

(A) Zn, Cu, Ag, Au (B) Zn, Ag, Au

(C) Au only (D) Cu, Ag, Au

Q37. Which process of purification is represented by the following scheme ?

$$\underset{\text{Impure}}{Ti} + 2I_2 \xrightarrow{250^\circ C} TI_4 \xrightarrow{1400^\circ C} \underset{\text{Pure}}{Ti} + 2I_2$$

(A) Cupellation

(B) Poling

(C) Electrolytic refining

(D) Van Arkel process

Q38. Impure Gold + $H_2O \xrightarrow[O_2]{KCN}$ (X);

(X) $\xrightarrow{Zn}$ (Y) + Au. 'X' and 'Y' are :-

(A) $K[Au(CN)_2]$, $K^+[Zn(CN)_4]^-$

(B) $[Au(CN)_2]$, $[Zn(CN)_4]^{-2}$

(C) $[Au(CN)_2]$, $[K(CN)_4]^{-2}$

(D) None of these

Extraction of Tin and Copper

Q39. Identify the statement which is not correct for copper sulphate :

(A) It reacts with NaOH and glucose to give Cu_2O

(B) It reacts with KCl to give Cu_2Cl_2

(C) It gives CuO on strong heating

(D) It reacts with KI to give I_2

Q40. Which of the following is extracted by air reduction.

(A) Graphite (B) Silicon

(C) Tin (D) Lead

Q41. Copper is extracted from its ore by-

(A) Electrolytic reductionn

(B) Auto reduction

(C) Cyanide process

(D) Magnetic separation

Q42. Extraction of zinc from zinc blende is achieved by :

(A) electrolytic reduction

(B) roasting followed by reduction with carbon

(C) roasting followed by self-reduction

(D) roasting followed by reduction with another metal

Q43. Which is not the ore of copper –

(A) Ruby copper

(B) Malachhite green

(C) Azurite

(D) Siderite

Q44. The slag obtained during the extraction of copper from copper pyrites is composed mainly of:-

(A) $CaSiO_3$ (B) $FeSiO_3$

(C) $CuSiO_3$ (D) SiO_2

Q45. Which of the following can perform as a basic flux in slag formation :-

(A) SiO_2 (B) CO_2

(C) FeO (D) $CaSiO_3$

Q46. Concentration of copper pyrites is done by the following method :-

(A) Magnetic separation

(B) Gravity separation

(C) Froth floatation process

(D) None

Q47. Which of the following step can not be used to the extraction of blister Cu from $CuFeS_2$

(A) Smelting

(B) Roasting

(C) Auto/Self reduction

(D) Poling

Q48. Which of the following pairs of metals is purified by van Arkel method ?

(A) Ga and In (B) Zr and Ti

(C) Ag and Au (D) Ni and Fe

Q49. Reduction in the metallurgy of copper pyrites takes place :-

(A) Hall Heroult's

(B) Alumino thermic process

(C) Smelting

(D) Bessemerisation

Extraction of Lead and Magnesium

Q50. Which one is ore of Mg -

(A) Anglesite (B) Dolomite

(C) Cerussite (D) Azurite

Q51. Of the following metals the one which cannot be obtained by electrolysis of the aqueous solution of the salt-

(A) Ag (B) Cu

(C) Mg (D) Au

Extraction of Iron

Q52. A substance which reacts with gangue to form fusible material is called-

(A) Flux (B) Catalyst

(C) Ore (D) Slag

Q53. The function of flux during the smelting of the ore is :

(A) To make the ore porous

(B) To remove gangue

(C) To facilitate reduction

(D) To facilitate oxidation

Q54. In steel making the impurities are removed by-

(A) Oxidation

(B) Reduction

(C) Converted into high melting compounds by addition of some elements

(D) Gasification

Extraction of Aliminium

Q55. Aluminium is obtained from Al_2O_3 by this method-

(A) Thermal reduction

(B) Hydro metallurgical method

(C) Electrolytic reduction

(D) Reduction by Iron

Q56. By which process pure aluminia is produced when bauxite is rich in silica.

(A) Hall's process

(B) Bayer's proces

(C) Serpeck's process

(D) (A) and (B) both

Q57. Which of the following is not used for extraction of aluminium-
(A) Van arkel process
(B) Serpeck's process
(C) Baeyer's process
(D) Hall-Heroult's process

Q58. Metal oxide is reduced to metal when :
(A) Calcinated calamine is treated with carbon
(B) Alumina reacts with iron
(C) Unroasted galena is heated with PbS
(D) Impure Ni is heated with carbon mono oxide

Q59. Aluminium metal is purified by :-
(A) Hoope's process
(B) Hall's process
(C) Sepeck's process
(D) Baeyer's process

Q60. Main ore of Aluminium in which amount of Al is very high :-
(A) Bauxite
(B) Beryl
(C) Gibbsite
(D) All of these

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

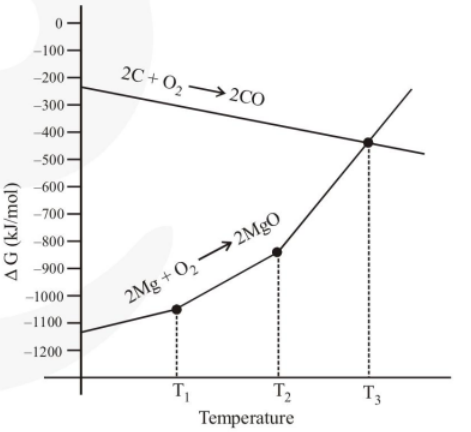
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question*

Q1. Choose the correct statements.
Roasting is carried out to:
(A) convert sulphide to oxide and sulphate
(B) remove water of hydration
(C) melt the ore
(D) remove arsenic and sulphur impurities

Q2. NaCl and $CaCl_2$ are also added to fused $MgCl_2$ in the electrolysis of $MgCl_2$. Select the incorrect statement(s) for this solution :
(A) Melting point is decreased and conductivity is increased.
(B) Melting point is increased and conductivity is decreased.
(C) Melting point and conductivity both are decreased.
(D) Melting point and conductivity both are increased.

Q3. Which is/are the correct method - metal matching for refining of crude metals?
(A) Distillation : zinc and mercury
(B) Liquation : tin
(C) Van Arkel : zirconium
(D) Mond process : lead

Q4. Correct statements about the plot is/are –



(A) T_1 and T_2 are melting point and boiling point of Mg respectively
(B) T_1 and T_2 are melting point and boiling point of MgO respectively
(C) Reduction of MgO by coke is possible above T_3
(D) Mg can be extracted from gaseous products by rapid cooling

Q5. (Ag + Pb) alloy $\xrightarrow[\text{is added}]{\text{melt and zinc}}$ (Ag + Pb + Zn) melt $\xrightarrow{\text{Cool}}$ $\frac{\text{Layer X}}{\text{Layer Y}}$

Select the correct statement(s) based on above scheme :

(A) Layer X contains zinc and silver.

(B) Layer Y contains lead and silver but amount of silver in this layer is smaller than in the layer X.

(C) X and Y are immiscible layers.

(D) X and Y are miscible layers.

Q6. Carbon can be used to reduce a number of oxides and other compounds, and because of the low cost and availability of coke, this method is widely used, but the disadvantages are that :

(A) High temperature is needed, which is expensive

(B) Many metals combine with carbon to form carbides

(C) Low temperature is needed, which is expensive

(D) Many metal combine with carbon and do not form carbides

Q7. In the leaching of Ag_2S with NaCN, a stream of air is also passed. It is because of :

(A) Reversible nature of reaction between Ag_2S and NaCN

(B) Oxidized Na_2S formed into Na_2SO_4

(C) Irreversible nature of reaction between Ag_2S and NaCN

(D) None of the above

Q8. Silver containing lead as an impurity is not purified by :

(A) Poling (B) Cupellation

(C) Levigation (D) Distillation

Q9. For the pyrometallurgical method used for the extraction of copper from sulphide ore, which statement(s) is/are correct?

(A) Pyrometallurgy is dry method

(B) It involves concentration by leaching the sulphide ore with dil. H_2SO_4

(C) It involves concentration of the sulphide ore by froth floatation process

(D) It involves concentration by leaching for every ore

Q10. SnO_2 is reduced to metallic Sn on smelting oxide with anthracite and lime. In this, the function of lime is :

(A) It acts as a flux

(B) It removes acidic impurities as slag

(C) It removes basic impurities as slag

(D) It acts as a slag

Q11. Which of the following is (are) regarded as iron ores –

(A) Haematite (B) Magnetite

(C) Limonite (D) Copper pyrites

Q12. Which of the following statment are correct?

(A) The process of producing a hard coating of iron nitride on the surface of steel is called nitriding.

(B) The process of producing a thin coating of iron carbide on the surface of mild steel is called case - hardening.

(C) Quenched steel is produced by heating steel to redness and allowing it to cool slowly.

(D) Stainless steel is produced by heating wrought iron in molten chromium.

Q13. Which of the following process(es) are used for purification of Bauxite ore –

(A) Hall's process

(B) Serpeck's process

(C) Baeyer's process

(D) Mond's process

Q14. The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is-

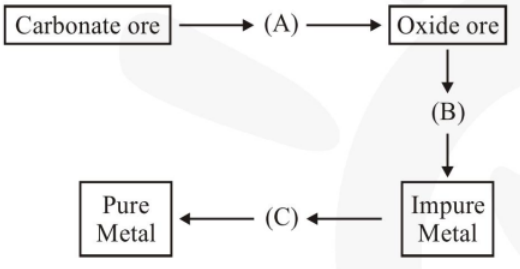
(A) As a catalyst
(B) To make the fused mixture very conducting
(C) To lower the temperature of the melt
(D) To decrease the rate of oxidation of carbon at aode.

Q15. In which of the following pair(s), the minerals are converted into metals by self-reduction process–
(A) Cu_2S, PbS (B) PbS, HgS
(C) PbS, ZnS (D) Ag_2S, Cu_2S

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph Question No.16 to 18



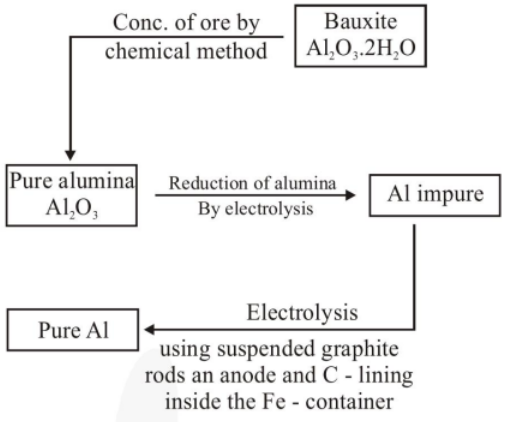
Q16. A metal is in combined state as carbonate; find out process A:
(A) Roasting (B) Calcination
(C) Smelting (D) Reduction

Q17. Find out process B :
(A) Roasting (B) Reduction
(C) Van Arkel (D) Smelting

Q18. Find out process C :
(A) Electrolytic refining
(B) Reduction
(C) Smelting
(D) Roasting

Paragraph Question No. 19 to 22

Extraction of aluminium can be understood by :



Electrolytic reduction of Al_2O_3 :
Electrolyte : (Al_2O_3 + Cryolite + CaF_2)
Cathode : Carbon inside the Fe - container
Anode : Graphite rods

Q19. The purpose of adding cryolite is :
(A) To decrease the eletrical conductivity of pure aluminium
(B) To lower the melting point of Al_2O_3
(C) To remove the impurities as slag
(D) To increase the Al percentage in the yield

Q20. Coke powder is spread over the molten electrolyte to :
(A) Prevent the heat radiation from the surface
(B) Prevent the corrosion of graphite anode
(C) Prevent the oxidation of molten alumnium by air
(D) Both (a) and (b)

Q21. The function of fluorspar (CuF_2) is :
(A) To increase the melting point of electrolyte
(B) To increase electrolytic coductivity power
(C) To remove the impurities as slag
(D) All of these

Q22. The molten eletrolytes contain Na^+, Al^{3+}, and Ca^{2+} but only Al gets deposited at cathode because:

(A) Standard reduction potential of Al is more than that of Na and Ca

(B) Standard oxidation potential of Al is more than that of Na and Ca

(C) Discharge potential of Al^{3+} is higher than Na^+ and Ca^{2+}

(D) Graphite reacts only with Al^{3+} and not with Na^+ and Ca^{2+}

Q23. Find the number of reaction from the given reactions which can show calcination process :

(A) $CaCO_3.MgCO_3 \xrightarrow{\Delta} CaO + MgO + CO_2$

(B) $CuCO_3.Cu(OH)_2 \xrightarrow{\Delta} 2CuO + H_2O + CO_2$

(C) $Al_2O_3.2H_2O \xrightarrow{\Delta} Al_2O_3 + 2H_2O$

(D) $2Cu_2S + 3O_2 \xrightarrow{\Delta} 2Cu_2O + 2SO_2$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q24. Match the column (for electrolytic refining of Cu) :

Column-I	**Column-II**
(A) Anode	(P) Thin sheets of pure Cu
(B) Cathode	(Q) An aqueous solution of copper sulphate containing H_2SO_4
(C) Electrolyte	(R) Ag, Au
(D) Anode mud	(S) Impure metal of Cu

Q25. Match the column –

Column-I	**Column-II**
(A) van Arkel method	(P) Refining of silver
(B) Solvay process	(Q) Purification of titanium
(C) Cupellation	(R) Manufacture of Na_2CO_3
(D) Poling	(S) Purification of copper

Q26. Match the column correctly –

Column-I	**Column-II**
(A) Cyanide process	(P) Extraction of Al
(B) Self reduction	(Q) Extraction of Ag
(C) Electrolytic reduction	(R) Extraction of Pb
(D) Carbon reduction	(S) Extraction of Sn

Q27. Match the conversions in Column I with the type(s) of reaction(s) given in Column II.

Column I	Column II
(A) $PbS \rightarrow PbO$	(P) roasting
(B) $CaCO_3 \rightarrow CaO$	(Q) calcination
(C) $ZnS \rightarrow Zn$	(R) Carbon reduction
(D) $Cu_2S \rightarrow Cu$	(S) self reduction

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q28. Find the number of ores, which can be concentrated by magnetic separation method.

Galena; Copper pyrites; Haematite; Siderite

Q29. Find the number of impurities which deposited as anode mud in the electrorefining of copper.

Antimony; Selenium; Tellurium; Silver; Gold; Platinum

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q30. **Statement 1 :** Wolframite impurities are separated from cassiterite by electromagnetic separation.

Statement 2 : Cassiterite being magnetic is attracted by the magnet and forms a separate heap.

Q31. **Statement 1 :** Silver ores and native gold have to be leached with metal cyanides.

Statement 2 : Silver and gold do not form complex ion with cyanide ion.

Q32. **Statement 1 :** Native Ag and Ag_2S are dissolved in CN^- solution in presence of Air (O_2).

Statement 2 : Air (O_2) acts as oxidant for Ag and Ag_2S both.

Q33. **Statement 1 :** Carbon reduction for Al_2O_3, is very difficult.

Statement 2 : Al forms carbide the Al_4C_3 which produces propyne when reacts with H_2O.

Q34. **Statement 1 :** Limestone is added to the blast furnace in the extraction of iron from haematite.

Statement 2 : Limestone at higher temperature give CaO (lime).

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Which of the following is not a concentration technique-

(A) Levigation

(B) Froth floatation

(C) Leaching

(D) Calcination

Q2. Which one of the following ores is best concentrated by froth-floatation method :-

(A) Magnetite (B) Cassitertite

(C) Galena (D) Malachite

Q3. $Ag_2S + NaCN \longrightarrow A$, $A + Zn \longrightarrow B + Ag$

A and B are respectively :

(A) $Na_3[Ag(CN)_3]$, $Na_2[Zn(CN)_4]$

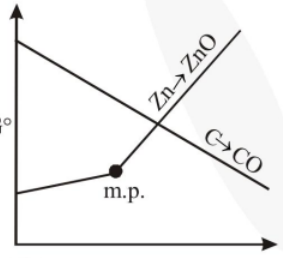
(B) $Na[Ag(CN)_2]$, $Na[Zn(CN)_3]$

(C) $Na_2[Zn(CN)_4]$, $Na_2[Ag(CN)_4]$

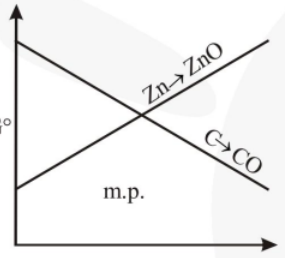
(D) $Na[Ag(CN)_2]$, $Na_2[Zn(CN)_4]$

Q4. Metallic zinc is collected in the form of vapour by the carbon reduction process, then the correct diagram for this phenomenon is/ are -

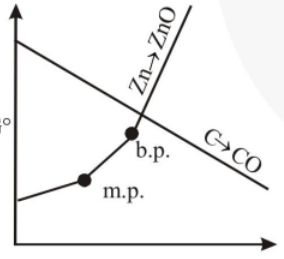
(A) $\Delta G°$



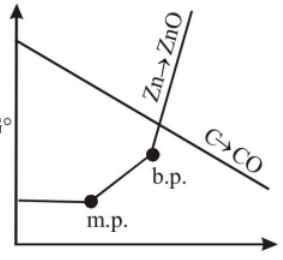
(B) $\Delta G°$



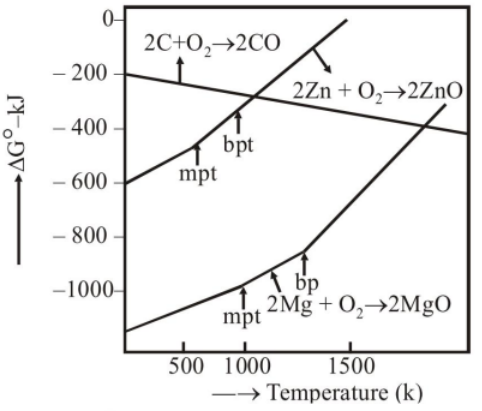
(C) $\Delta G°$



(D) $\Delta G°$



Q5. At 1100°C, which reaction is spontaneous to a maximum extent ?



(A) $MgO + C \rightarrow Mg + CO$

(B) $ZnO + C \rightarrow Zn + CO$

(C) $MgO + Zn \rightarrow Mg + ZnO$

(D) $ZnO + Mg \rightarrow Zn + MgO$

Q6. In the extraction of silver, Ag_2S is dissolved in -

(A) HCl (B) HNO_3

(C) KCN (D) H_2SO_4

Q7. Ag can be separated from Pb by :

(A) Fractional crystallisation

(B) Amalgamation

(C) Filtration

(D) Addition of Zn

Q8. To obtain silver from silver amalgam it is heated in vessel which is made of –

(A) Cu (B) Fe (C) Ni (D) Zn

Q9. Which of the following pairs of metals is purified by van Arkel method ?

(A) Ga and In (B) Zr and Ti

(C) Ag and Au (D) Ni and Fe

Q10. Removal of Fe, Cu, W from Sn metal after smelting is done by ..... because .....

(A) Poling; of more affinity towards oxygen for impurities.

(B) Selective oxidation; of more affinity towards oxygen for impurities.

(C) Electrolytic refining; impurities undissolved in electrolyte

(D) Liquation; Sn having low melting point compared to impurities.

Q11. Bell metal is an alloy of :

(A) Cu + Pb (B) Cu + Sn

(C) Cu + Zn (D) Cu + Ni

Q12. The flux used in the smelting of copper ores is :

(A) Lime stone (B) Magnesia

(C) Silica (D) Coke

Q13. In the extraction of Cu, the metal is formed in the bessemer converter due to the reaction:

(A) $Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$

(B) $Cu_2S \longrightarrow 2Cu + S$

(C) $Fe + Cu_2O \longrightarrow 2Cu + FeO$

(D) $2Cu_2O \longrightarrow 4Cu + O_2$

Q14. In the extraction of Cu from copper pyrites, iron is removed as :

(A) $FeSO_4$ (B) $FeSiO_3$

(C) Fe_3O_4 (D) Fe_2O_3

Q15. Ore (of metal **M**) $\xrightarrow{\text{Calcination}}$ Residue $\xrightarrow{\text{dil } HNO_3}$ aq. solution $\xrightarrow{\text{Zn dust}}$ Metal **(M)** + Zn^{2+}(aq.)

above metallurgy is possible when ore is -

(A) $ZnCO_3$

(B) $CaCO_3.MgCO_3$

(C) $CuCO_3.Cu(OH)_2$

(D) PbS

Q16. The final step for the extraction of copper from copper pyrities in bessemere converter involves the reaction :-

(A) $4Cu_2O + FeS \rightarrow 8Cu + FeSO_4$

(B) $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$

(C) $2Cu_2O + FeS \rightarrow 4Cu + Fe + SO_2$

(D) $Cu_2S + 2FeO \rightarrow 2Cu + 2FeCO + SO_2$

Q17. Which of the following step is not involved in the extraction of blister copper from copper pyrite.

(A) Smelting

(B) Roasting

(C) Auto/Self reduction

(D) Polling

Q18. Select the incorrect one :-

(A) Blister copper has impurity of 98%

(B) Van Arkel is used for purification of Zr.

(C) Al_2O_3 can not be reduced by Gold schmit alumino thermic process.

(D) NaCN is used as deprassant in froth floatation process.

Q19. Principle in the dow's process is-

(A) MgO is reduced with 'C' in clay retorts

(B) $MgCl_2$ electrolysis

(C) MgO is reduced with Fe/Si in clay retorts

(D) None of these

Q20. Consider the following statements :

(A) When the lead-silver alloy rich in silver, lead is removed by the cupellation process

(B) Froth flotation can be applied for non sulphide ore also using suitable activator

(C) $PbSO_4 + H_2SO_4$, electrolyte is used for the refining of Pb by electrolysis

(D) Any metal will not reduce the oxide of other metals which lie above it in the Ellingham diagram.

Using 'T' for true and 'F' for false statements in the given sequence, select the correct set of code:

(A) T T F T (B) T T F F

(C) F T F T (D) T T T F

Q21. On igniting Fe_2O_3 at 1400°C, in presence of carbon the product obtained is-

(A) Fe_2O_3 melt (B) FeO

(C) Fe_3O_4 (D) Metallic iron

Q22. Which of the following metals burns in air at high temperature with the evolution of much heat ?

(A) Cu (B) Hg (C) Pb (D) Al

Q23. The purpose of smelting of an ore is

(A) to oxidise it

(B) to reduce it

(C) To remove volatile impurities from it

(D) to obtain an alloy

Q24. Which of the following metals will not form an amalgam ?

(A) Gold (B) Silver

(C) Zinc (D) Iron

Q25. In Fe-extraction, the roasting is adopted though the ore is not having any sulphide because -

(A) Haematite is to be decomposed

(B) All FeO is to be converted into Fe_2O_3

(C) All Fe_2O_3 is to be converted into FeO

(D) Slag formation is encouraged

Q26. Which of the following ore is calcinated during extraction :-

(a) Argentite (b) Siderite

(c) Azurite (d) Limonite

(A) a & d (B) b & c

(C) b, c & d (D) a, c & d

Q27. Select process in which SO_2 is **NOT** evolved:

(A) When Cu_2S is roasted

(B) When Cu_2S is heated with roasted Cu_2S

(C) When FeS is partially roasted

(D) When partially roasted FeS is heated with Silica

Q28. Which of the following match are incorrect:-

(A) Goldschmidt aluminothermite process - Cr_2O_3

(B) Mac Arther cyanide process - Fe

(C) Mond process - Ni

(D) Van Arkel process – Au

(A) A, C (B) C, D

(C) B, D (D) A, B

Q29. The element which could be extracted by electrolytic reduction of its oxide dissolved in a high temperature melt is-

(A) Sodium

(B) Magnesium

(C) Fluorine

(D) Aluminium

Q30. In the extraction of Al, the function of cryolite is to :

(A) Lower the melting point of alumina

(B) Increase the melting point of alumina

(C) Remove impurities from alumina

(D) Minimise the anodic effect

Q31. The mass of carbon anode consumed (giving only carbondioxide) in the production of 270 kg of aluminium metal from bauxite by the Hall process is (Atomic mass Al = 27)

(A) 180 kg (B) 270 kg

(C) 540 kg (D) 90 kg

Q32. Select the correct option :-

(A) Auto reduction ⇒ Example Ca(Calcium)

(B) Mond's process ⇒ Example Ti

(C) Hydrometallurgy ⇒ Example Al

(D) Zone refining ⇒ For semiconductor

Q33. Which of the following correctly matched

(A) Monds process - Ni

(B) Zone Refining - Ti

(C) Hoop's Method - Al

(D) Hydro metallurgy - Ag

Correct code is :-

(A) A, B, D

(B) B, C, D

(C) A, C, D

(D) A, B, C, D

Q34. In the extraction of aluminium.
Process X : applied for red bauxite to remove iron oxide (chief impurity)
Process Y : (Serpeck's process) : applied for white bauxite to remove Z (chief impurity) then, process X and impurity Z are
(A) X = Hall and Heroult's process and Z = SiO_2
(B) X = Baeyer's process and Z = SiO_2
(C) X = Serpeck's process and Z = iron oxide
(D) X = Baeyer's process and Z = iron oxide

Q35. Regarding electrolysis of Al_2O_3 by hall heroult process, which statement is false ?
(A) Cryolite Na_3AlF_6 lowers the melting point and increases the electrical conductivity of Al_2O_3
(B) Al is obtained at cathode and probably CO_2 at anode
(C) Electrolysis is carried out in aqueous medium
(D) Anode is made of graphite

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following ores is (are) concentrated by froth floatation -
(A) haematite (B) galena
(C) copper pyrite (D) azurite

Q2. Select the correct statement(s) ;
(A) In hydrometallurgy, Zn is used as oxidizing agent in the purification of Ag from $[Ag(CN)_2]^-$
(B) When pine oil or eucalyptus oil is added into the water, it lowers down the surface tension by which froth is formed.
(C) Sodium ethyl xanthate is used as collector.
(D) Basic copper carbonate or $PbSO_4$ is concentrated by froth floatation method by using and activator.

Q3. Which statement is incorrect regarding thermite process –
(A) 3 part metaloxide + 1 part Al is taken as thermite mixture.
(B Ignition mixture (Mg powder + BaO_2) acts as catalyst.
(C) Metals (Fe, Cr) are obtained in vapour form
(D) Reaction is exothermic due to highly +ve enthalpy of formation of Al_2O_3.

Q4. Choose the correct options –

$$C_{(s)} + \frac{1}{2}O_2 \longrightarrow CO_{(g)},\ \Delta H^\circ = -26\ \text{K.cal mol}^{-1}$$

$$CO_{(g)} + \frac{1}{2}O_2 \longrightarrow CO_{2(g)},\ \Delta H^\circ = -68\ \text{K.cal mol}^{-1}$$

(A) Heat of formation of CO_2 is - 68 K.cal mol^{-1}.
(B) Heat of combustion of $C_{(s)}$ is - 26 K.cal mol^{-1}
(C) Heat of combustion of $CO_{(g)}$ is - 68 K.cal mol^{-1}
(D) Heat of formation of $CO_{(g)}$ is - 26 K.cal mol^{-1}

Q5. Select the correct statement(s) :
(A) When temperature is raised, a point will be reached where the graph crosses the $\Delta G = 0$ line. Below this temperature, the free energy of formation of the oxide is negative, so the oxide is stable.
(B) When temperature is raised, a point will be reached where the graph crosses the $\Delta G = 0$ line. Above this temperature, the free energy of formation of the oxide is positive, and the oxide becomes unstable, and should decompose into the metal and dioxygen.

(C) Theoretically, all oxides can be decomposed to give the metal and dioxygen if a sufficiently high temperature can be attained.

(D) Theoretically, all oxides cannot be decomposed to give the metal and dioxygen if a sufficiently high temp. can be attained.

Q6. Extraction of silver from argentiferrous lead (Pb + Ag) involves –

(A) distillation method

(B) cupellation

(C) froth flotation method

(D) treatment with NaCl

Q7. For the extraction of Pb when impurity content is very high, then during roasting of galena, the formation of which compound (s) is/are prevented by lime –

(A) $PbSO_4$ (B) PbO_2

(C) $PbSiO_3$ (D) $CaSiO_3$

Q8. Auto - reduction process is used for the extraction of :

(A) Cu (B) Hg

(C) Pb (D) Al

Q9. Which of the following statement(s) is (are) true –

(A) In process of the precipitation of silver from sodium dicyano argentate (I), the zinc acts as reducing agent as well as complexing agent.

(B) In process of the roasting, the copper pyrites is converted into a mixture of Cu_2S and FeS which, in turn, are partially oxidised.

(C) Limonite, haematite and magnesites are ores of iron

(D) Tin and lead both are extracted from their ores by self-reduction.

Q10. Hoop's process of purification of aluminium involves formation of layers during electrolysis. It involves –

(A) The three layers have same densities but different materials

(B) The three layers have different densities

(C) The upper layer is of pure aluminium which acts as a cathode

(D) The bottom layer is of impure aluminium which acts as an anode and middle layer consists of cryolite and BaF_2

Q11. Which of the following reactions are actually employed in commercial extraction of metals ?

(A) $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$

(B) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$

(C) $2Na[Au(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Au$

(D) $Cu_2S + Pb \rightarrow Cu + PbS \downarrow$

Q12. Metals which can be extracted by smelting process is/are:

(A) Sn (B) Cu

(C) Zn (D) Al

Q13. In Baeyer's process, which of the following statements are incorrect ?

(A) Al_2O_3 goes into solution as soluble $Al(OH)_4^-$ while other basic oxides as TiO_2 and Fe_2O_3 remain insoluble.

(B) Al_2O_3 changes to AIN which in turn is decomposed by H_2O.

(C) Al_2O_3 changes to $Al_2(CO_3)_3$ which changes to $AlCl_3$.

(D) Al_2O_3 changes to $Al_2(SO_4)_3$ which changes to Al_2S_3.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question 14 to 16

Castner-Keller cell is the common cell in which mercury is used as cathode. The advantage of using Hg as a cathode is that the discharge potential of Na^+ ions is less than that of H^+ ions. Na^+ ions get discharged on mercury and sodium so deposited combines with mercury to form sodium amalgam. The cell consists of a large rectangular through divided into three compartments by slate partitions which do not touch the bottom of the cell but dipping in mercury.

When electricity is circulated, sodium chloride in the outer compartments is electrolysed. Chlorine is evolved at the graphite anode white Na^+ ions are discharged at the Hg cathode. The liberated sodium forms amalgam with mercury.

$$NaCl \square Na^+ + Cl^-$$

At anode $2Cl^- \longrightarrow 2Cl + 2e \longrightarrow Cl_2$

At cathode $Na^+ + e \longrightarrow Na$

$Na + Hg \longrightarrow$ Amalgam

The sodium amalgam thus formed comes in the inner compartment due to rocking. Here, the sodium amalgam acts as the anode and iron rods acts as cathode.

An anode Na-amalgam $\longrightarrow Na^+ + Hg + e$

At cathode $2H_2O + 3e \longrightarrow H_2 \uparrow + 2OH^-$

Q14. In Castner-Keller cell, anode is -

(A) Graphite (B) amalgam

(C) Both A and B (D) Mercury

Q15. The NaOH is produced in :

(A) Sider compartment

(B) Centre compartment

(C) Both

(D) NaOH is not produced

Q16. The wheel is used in Castner-Keller cell is useful to help in-

(A) Transfer of amalgam from sider compartment to centre compartment.

(B) Transfer of Hg from centre compartment to sider compartment.

(C) Both A and B

(D) None of these

Paragraph for Question 17 to 20

Lead ontained from galena ore (PbS) by air reduction or carbon reduction process contains base metal (Cu, Bi, Sn, As) as impurities; it is due to the presence of these impurities that lead becomes hard and brittle.

Q17. Parke's process is also called :

(A) Softening process

(B) Desilverisation method

(C) Cupellation

(D) None of these

Q18. Zn - Ag alloy formed in the upper layer of molten lead is skimmed off from the surface of the molten lead by perforated ladles. This alloy contains lead as impurity. This impurity of Pb is removed by :

(A) Distillation process

(B) Cupellation

(C) Liquation

(D) Bett's electrolysis

Q19. Ag can be obtained from purified Zn - Ag alloy by :

(A) Distillation (B) Poling

(C) Liquation (D) Reduction

Q20. Cupellation is used from for purification of:

(A) Pb (B) Ag

(C) Zn (D) Fe

Paragraph for Question 21 to 23

Amongst the various ores of a metal (M) (sulphide, carbonates, oxides, hydrated or hydroxides) two ores [X] and [Y] show the following reactivity.

(i) [X] on calcination gives a black solid (S), carbon dioxide and water.

(ii) [X] Dissolved in dil. HCl on reaction with KI gives a white ppt. (P) and iodine.

(iii) [Y] on roasting gives metal (M) and a gas (G_1) which turns acidified $K_2Cr_2O_7$ solution green.

(iv) [Y] on reaction with dil. HCl gives a white ppt. (MS) and another gas (G_2) which turns lead acetate solution black and also reacts with gas (G_1) to precipitate colloidal sulphur in presence of moisture.

The MS, [X] and [Y] gives greenish blue flame.

Q21. The metal ores [X] and [Y] are respectively:
(A) Carbonate and sulphide ores
(B) Sulphide and carbonate ores
(C) Carbonate and hydroxide ores
(D) Carbonate and oxide ores

Q22. Which of the following statements is correct about [Y] ?
(A) [Y] is converted to metal (M) by self reduction.
(B) Carbonate extract of [Y] gives yellow ppt with suspension of $CdCO_3$.
(C) [Y] is chalcocites or chalcopyrites
(D) All of these

Q23. The gas (G_1) acts as
(A) Oxidising agent
(B) reducing agent
(C) oxidising and reducing agent
(D) fluxing agent

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q24. Match the column

Column-I (Commercial methods)	**Column-II (Metals)**
(A) $MS + 2O_2 \xrightarrow{\Delta} MSO_4$ $MSO_4 + MS \xrightarrow{\Delta} 2M + 2SO_2$	(P) Ag
(B) The metal(s) (during extraction of metal) is/are not reduced during smelting step	(Q) Au
(C) The ore of the metal(s) is/are concentrated by froth floatation method	(R) Cu
(D) Impure metal + CN^- soluiton → soluble complex Soluble complex + Zn → pure metal	(S) Pb

Q25. Match the column :

Column-I	**Column-II**
(A) Bauxite ore	(P) Roasting
(B) SnO_2 ore	(Q) Calcination
(C) Native silver ore	(R) Leaching
(D) $CuFeS_2$ ore	(S) Electrolysis

Q26. Match the column –

Column-I	**Column-II**
(A) Iron and copper	(p) Poling
(B) Zirconium and Titanium	(q) Bessemerisation
(C) Lead and Tin	(r) Van-Arkel
(D) Copper and Tin	(s) Liquation

Q27. Match the items of Column I with items of Column II and assign the correct code :

Column I	**Column II**
(A) Blistered Cu	(p) Aluminium
(B) Blast furnace	(q) $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$
(C) Reverberatory furnace	(r) Iron
(D) Hall-Heroult process	(s) $FeO + SiO_2 \rightarrow FeSiO_3$
	(t) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

Code :

(A) A → (q) ; B→ (r) ; C→ (s) ; D → (p) (B) A→ (p) ; B → (q) ; C → (r) ; D→ (t)

(C) A → (t) ; B→ (s) ; C → (r) ; D → (q) (D) A → (s) ; B → (t) ; C→ (r) ; D → (q)

Q28. Match items of Column - I with the items of Column - II and assign the correct code :

Column-I	**Column - II**
(A) Cyanide process	(P) Ultrapure Ge
(B) Froth floatation process	(Q) Dressing of ZnS
(C) Electrolytic reduction	(R) Extraction of Al
(D) Zone refining	(S) Extraction of Au
	(T) Purification of Ni

Code :

(A) A → (p) ; B→ (q) ; C→ (r) ; D → (s) (B) A→ (q) ; B → (s) ; C → (t) ; D→ (r)

(C) A → (q) ; B→ (r) ; C → (s) ; D → (t) (D) A → (s) ; B → (t) ; C→ (r) ; D → (q)

TRIPLE MATCH COLUMN TYPE

*This section contains TWO tables (each having 3 columns and 4 rows). Based on each table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column-I (ORE)	**Column - II (CONCENTRATION)**	**Column - III (REFINING)**
(P) PbS	(I) Calcination	(i) Vapour phase Refining
(Q) $CuFeS_2$	(II) Roasting	(ii) Poling
(R) $ZnCO_3$	(III) Neither Calcination nor roasting	(iii) Parke's process
(S) $Al_2O_3.xH_2O$	(IV) Both calcination & Roasting	(iv) Electro refining

Q29. Which is a correct match :-

(A) P, I, (iii) (B) Q, II, (ii)

(C) R, IV, (i) (D) S, III, (iii)

Q30. Which is incorrect match :-

(A) P, II, (ii) (B) Q, II, (ii)

(C) R, I, (i) (D) S, III, (iv)

Q31. For which metal extraction, II & (iii) are correct match :-

(A) Pb (B) Cu

(C) Zn (D) Al

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q32. Find the number of metals which are commercially reduced by carbon reduction method from the given metals :

Ag, Cr, Mn, Sn, Zn, Fe

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q33. **Statement 1 :** Froth floatation method is used for benefication of galena through ZnS is present as impurity.

Statement 2 : NaCN or KCN solution is very good depressant to suppress the floating property of PbS.

Q34. **Statement 1 :** Copper and silver are found in the combined state as sulphides in nature.

Statement 2 : Copper and silver are below hydrogen in electrochemical series.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	B	C	C	A	A	B	A	D	D
11	12	13	14	15	16	17	18	19	20
B	B	C	D	D	A	B	C	A	D
21	22	23	24	25	26	27	28	29	30
D	A	C	A	C	D	A	D	B	D
31	32	33	34	35	36	37	38	39	40
D	B	B	A	C	C	D	D	B	D
41	42	43	44	45	46	47	48	49	50
B	B	D	B	C	C	D	B	D	B
51	52	53	54	55	56	57	58	59	60
C	A	B	A	C	C	A	A	A	A

Ex-1A

1. A, B, D **2.** B, C, D **3.** A, B, C **4.** A, C, D **5.** A, B, C

6. A, B **7.** A, B **8.** A, C, D **9.** A, C **10.** A, B

11. A, B, C **12.** A, B **13.** A, B, C **14.** B, C **15.** A, B

16. B **17.** B **18.** A **19.** B **20.** D

21. B **22.** A **23.** A,B,D

24. (A) → (S); (B) → (P); (C) → (Q); (D) → (R)

25. A - Q ; B - R ; C - P ; D - S

26. A - Q ; B - R ; C - P ; D - S

27. (A) → P, (B) → Q, (C) → P, R, (D) → P, S

28. 2 **29.** 6 **30.** C **31.** C **32.** C

33. C **34.** A

Ex-2

1	2	3	4	5	6	7	8	9	10
D	C	D	C	D	C	D	B	B	D
11	12	13	14	15	16	17	18	19	20
B	C	A	B	C	B	D	A	B	B
21	22	23	24	25	26	27	28	29	30
D	D	B	D	B	C	D	C	D	A
31	32	33	34	35					
D	D	C	B	C					

Ex-2A

1. B, C **2.** B, C, D **3.** B, C, D **4.** C, D **5.** A, B, C

6. A, B, **7.** A, C **8.** A, B, C **9.** A, B **10.** B, C, D

11. B, C **12.** A, B, C **13.** B, C, D **14.** A **15.** B

16. C **17.** B **18.** C **19.** A **20.** B

21. A **22.** D **23.** C

24. (A) - R, S, (B) - R, (C) - P, R, S, (D) - P, Q

25. (A) → (Q,R,S); (B) → (P,R,S); (C) → (R,S); (D) → (P,S)

26. A - Q, B - R, C - S, D - P **27.** A **28.** B **29.** B

30. A **31.** A **32.** 3 **33.** C **34.** B